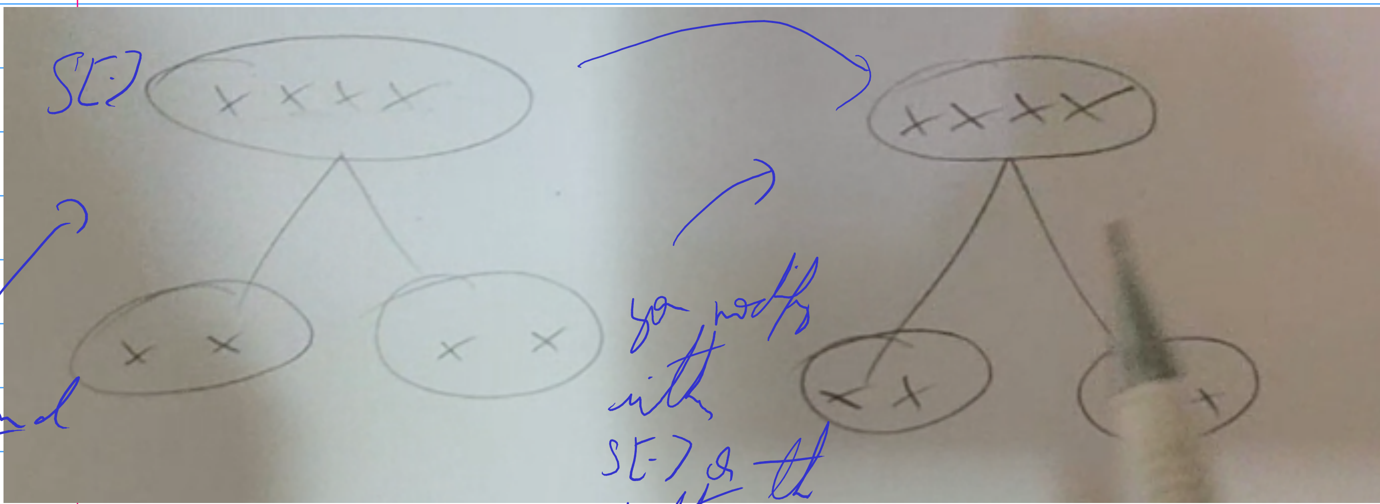
26/06/26

Dual-Tree DTWT -> DT WT WT

↳ pair almost invariant in time

↳ two decomposition trees



S[·]

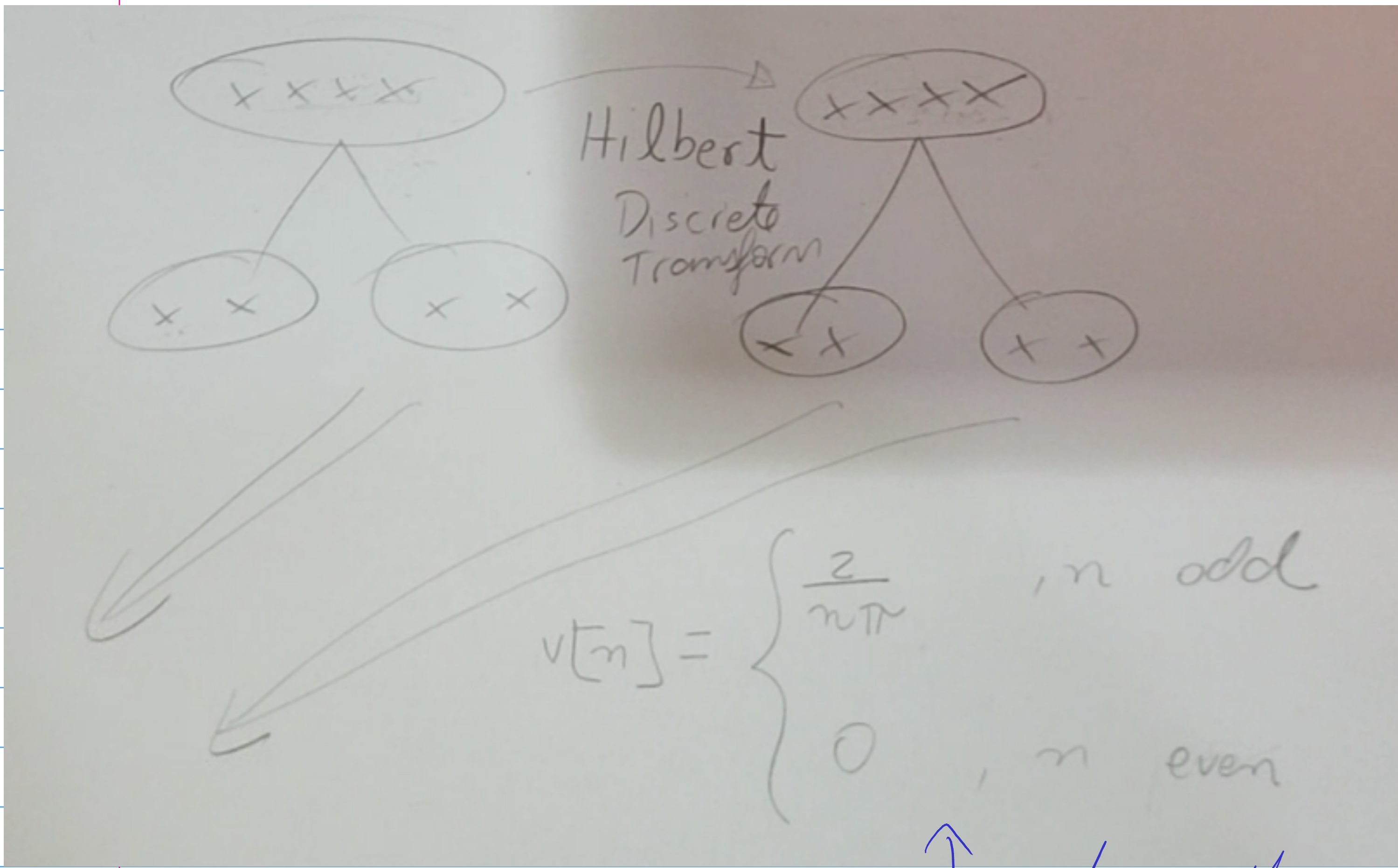
normal tree

you modify with S[-] & the filter

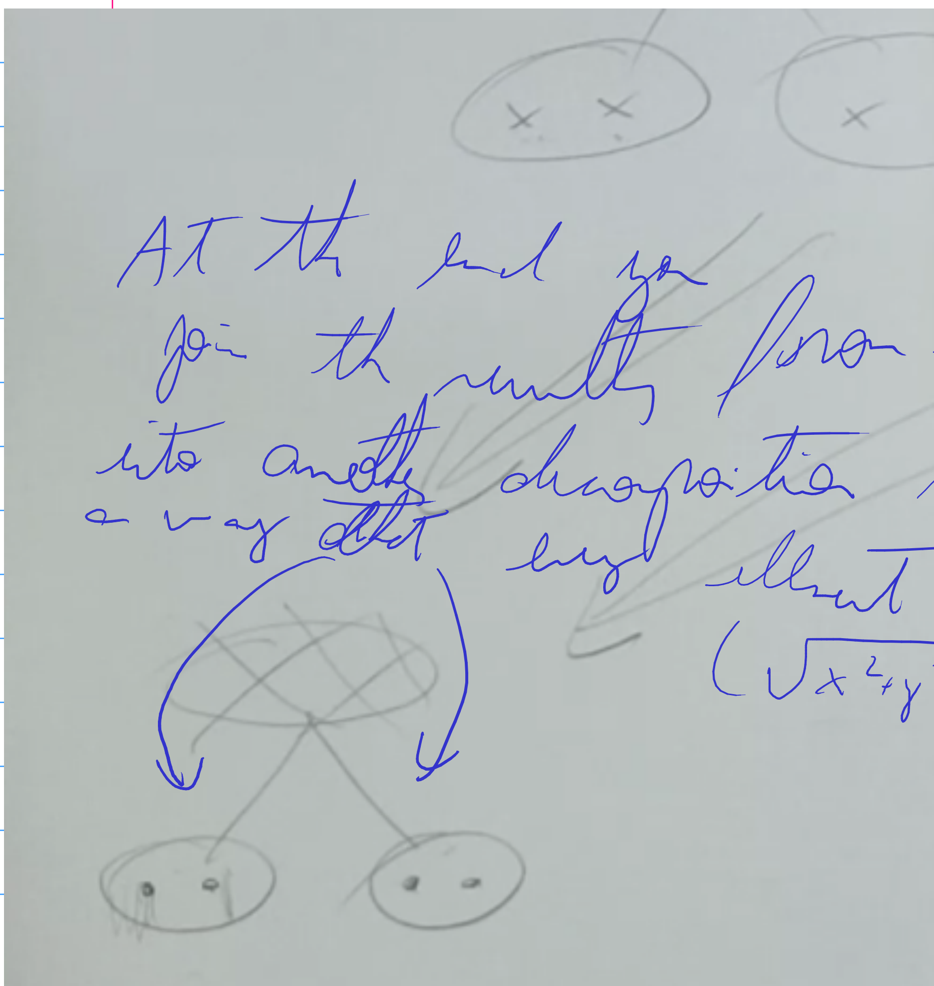
It's like if you had a complex value and a imaginary part

It's like Fourier

To map with the signal & filter to the related one we use Hilbert (inverse) transform (it's a filter of this type for all) to shift it by 90°



the filter for all



At the end you join the results from both trees into another decomposition tree in such a way that biggest element is the middle ($\sqrt{x^2+y^2}$)

it makes the decomposition be approximately stationary (invariant) without increasing the size as SDWT

Curvelet → for 2D
↳ search for curves
↳ matched / wavelets / tailored for pattern matching
↳ extract features / patterns
↳ of different sizes and angles

Denoise DTWT → DDTWT
↳ optimized for denoising acoustic signals
↳ usually to denoise, you modify the signal before inverting. In DDTWT before modifying, you adjust the value by a factor K that reflects the human auditory system. Later we apply 1/K

Matched wavelets → filters matched for an specific signal
↳ we map the signal into a plane Time - Frequency - Format

↳ it holds some format data

lifting scheme ∃ a way to separate DWT into a series of elementary convolution called lifting steps

If we unspan the main level you reach artifacts increasing the energy